

Evaluate the following integrals:

1. $\int \frac{2}{x-3} dx$

2. $\int \frac{2}{x-3} + \frac{5}{(x-3)^2} - \frac{1}{(x-3)^3} dx$

3. $\int \cos^2 x \sin x dx$

4. $\int \frac{5}{\sqrt{x^2+4}} dx$

5. $\int x e^{2x} dx$

6. $\int x e^{x^2} dx$

7. $\int \sin^2(3x) dx$

8. $\int \frac{3}{x^2+4x} dx$

9. $\int 2x \cos(3x) dx$

10. $\int \sqrt{5-x^2} dx$

11. $\int \frac{x}{x^2-25} dx$

$$12. \int e^{2x} \sin(5x) \, dx$$

$$13. \int \frac{2x}{x^2 + 6x + 5} \, dx$$

$$14. \int \frac{x - 1}{x^2 + 6x + 9} \, dx$$

$$15. \int \frac{1}{x^2 + 6x + 14} \, dx$$

$$16. \int \frac{1}{\sqrt{x^2 + 6x + 14}} \, dx$$

$$17. \int \frac{2x + 3}{x^2 + 6x + 14} \, dx$$

$$18. \int \frac{3 \sin x \, dx}{\cos^2 x + 9}$$

$$19. \int x^4 \sin(2x) \, dx$$

$$20. \int \frac{\sin(\ln 3x)}{2x} \, dx$$

$$21. \int \frac{4x + 2}{x^2 - x} \, dx$$

$$22. \int \frac{6x^2 - 7x}{3x - 2} \, dx$$